

Video Transcripts

Module 11: AI in Practice – Ecosystems and further advancements

Video 1: Business Ecosystems in the Modern World

In today's digital economy, competition increasingly takes place between business ecosystems rather than individual organisations. Modern products and services are rarely delivered by a single company; instead, they depend on networks of partners, platforms, technologies and service providers working together. A business ecosystem consists of interconnected participants that collectively create and deliver value, with each participant's performance influencing the success of the others.

A simple example is a food delivery application, where the customer experience depends on multiple contributors, including restaurants, payment providers, delivery partners, cloud services and customer support teams. This interconnected model is even more important in the era of Generative and Agentic AI. AI-powered solutions rely on foundation models, cloud infrastructure, data pipelines, enterprise systems and governance frameworks. What appears to be a single AI interaction is often the result of several organisations and technologies working together within a broader ecosystem.

Companies such as Google and Apple demonstrate different approaches to ecosystem strategy. Google operates an open ecosystem that encourages participation from developers, device manufacturers and enterprise customers, creating innovation through collaboration. Apple follows a more tightly controlled model, integrating hardware, software and services to deliver a seamless user experience. These examples highlight that ecosystems are not only about participation but also about how value is orchestrated and managed.

Three characteristics define successful business ecosystems: interdependence, network effects and orchestration. Participants rely on one another, creating shared value through collaboration. As more users and partners join, network effects strengthen the ecosystem and make it more valuable. At the same time, a central orchestrator often establishes the rules, standards and structures that guide ecosystem growth. In AI ecosystems, cloud providers and foundation model developers increasingly play this role by defining platforms, APIs and governance standards.

A key strategic challenge is balancing openness and control. Open ecosystems can encourage innovation and rapid adoption but may be harder to govern, while closed ecosystems offer greater consistency and control at the expense of flexibility. For organisations adopting Generative and Agentic AI, success depends not only on choosing the right technologies but also on participating in the right ecosystem, building effective partnerships and leveraging interconnected capabilities to create sustainable business value.

Video 2: Closed vs Open Ecosystems

Apple and Android represent two different approaches to building and managing business ecosystems. Apple follows a closed ecosystem model, controlling its hardware, operating system, application marketplace and digital services. This high level of integration allows Apple to deliver a

seamless and consistent user experience while maintaining strong oversight of security, performance and reliability. By controlling multiple layers of the ecosystem, Apple is also able to capture value across products, services and developer interactions.

In the context of Generative and Agentic AI, a closed ecosystem provides greater control over how AI capabilities are deployed and managed. AI features can be deeply integrated into devices and services, creating a more unified user experience while ensuring privacy and governance standards are maintained. However, this approach can limit the flexibility available to external developers, as innovation generally occurs within boundaries established by the ecosystem owner.

Android, by contrast, follows a more open and collaborative ecosystem model. While Google provides the core operating system and essential services, device manufacturers and developers have considerable freedom to customise products and build new applications. This openness has enabled Android to expand rapidly across global markets and support a wide range of devices. In AI applications, developers can experiment more freely and introduce innovative capabilities across different platforms, encouraging faster adoption and broader participation.

These two approaches highlight a fundamental strategic trade-off between control and flexibility. Closed ecosystems prioritise consistency, governance and value capture, while open ecosystems emphasise innovation, scale and experimentation. For organisations adopting Generative and Agentic AI, choosing the right ecosystem is a strategic decision that affects how quickly they can innovate, how effectively they can manage risk and how much value they can create over the long term.

Video 3: Platform & Super Platform Ecosystems

Platform ecosystems have become a central feature of the modern digital economy. Whenever people shop online, use cloud services, stream content or interact with digital applications, they are often participating in a platform ecosystem. These ecosystems connect different groups of users and enable value to be exchanged between them. As platforms expand across multiple services and industries, they can evolve into super platforms that create value on a much larger scale.

A platform ecosystem is built around multi-sided interactions between participants such as customers, developers, service providers and business partners. Xiaomi is a strong example of this model. Beyond smartphones, its ecosystem connects a wide range of devices, applications and IoT services through integrated platforms. Consumers, developers and device manufacturers all contribute to the ecosystem, while data generated across products helps improve personalisation, automation and user experiences. The more participants that join, the more valuable the ecosystem becomes for everyone involved.

Super platforms operate on a broader scale by integrating multiple industries and services within a single ecosystem. Companies such as Google and Amazon have expanded beyond their original offerings to include cloud computing, digital advertising, AI services, streaming platforms, productivity tools and e-commerce. These interconnected services create seamless experiences where users, businesses and developers can access multiple capabilities through a single ecosystem.

A defining characteristic of super platforms is their ability to create ecosystem spillover. Users often enter through one service and gradually adopt additional products and services within the same ecosystem. For example, a business using cloud services may later adopt AI tools, advertising solutions or productivity applications offered by the same provider. This interconnected structure strengthens customer engagement and creates self-reinforcing growth across multiple domains.

In the context of Generative and Agentic AI, platform and super platform ecosystems are becoming even more important. AI capabilities are increasingly integrated across services, enabling better coordination, automation and personalisation. Rather than operating within isolated applications, future AI systems will function across interconnected ecosystems where cloud infrastructure, commerce, data and intelligent services work together to deliver greater value for users and organisations.

Video 4: AI-Driven Shift in Competitive Dynamics

Artificial intelligence is fundamentally changing how organisations compete. In the past, businesses largely competed as individual firms, with success determined by factors such as product quality, pricing and operational efficiency. Today, AI-powered solutions rely on interconnected networks of data, models, cloud infrastructure and digital platforms. As a result, competition is increasingly shifting from firm versus firm to ecosystem versus ecosystem, where value is created through collaboration among multiple participants.

One of the most significant changes is the move towards ecosystem-level innovation. Modern AI applications often combine capabilities from several partners, including cloud providers, AI model developers, software vendors and domain specialists. This allows organisations to launch new products and services much faster than before. By leveraging shared platforms, APIs and infrastructure, businesses can focus on their unique strengths while benefiting from innovations developed across the broader ecosystem.

AI also enables the creation of shared intelligence through the aggregation of data across large networks of users and organisations. Platforms can analyse vast amounts of information to generate insights that individual participants could not produce on their own. As more users and partners contribute data, AI models become more accurate and valuable, creating self-reinforcing network effects that strengthen the overall ecosystem and enhance its competitive position.

Another important shift is the move from product differentiation to ecosystem differentiation. Organisations increasingly compete based on the strength of their partnerships, integrations and connected services rather than on standalone products alone. Companies that can combine multiple services, data sources and AI capabilities into a seamless experience often create greater value for customers and achieve stronger long-term engagement.

At the same time, AI is increasing dependence on shared platforms for critical resources such as data, computing power and foundation models. Businesses must balance the benefits of ecosystem participation with the risks of platform dependency and reduced control. Ultimately, success in the era of Generative and Agentic AI will depend not only on developing strong products but also on

effectively participating in, shaping and orchestrating intelligent ecosystems that drive innovation and sustainable competitive advantage.

Video 5: Ecosystems with a Dominant Focal Firm (Google Case Study)

Google is one of the most prominent examples of an ecosystem built around a dominant focal firm. While many participants contribute value, including device manufacturers, developers, advertisers and users, Google acts as the central orchestrator that connects and coordinates these activities. Through its portfolio of platforms and services, Google influences how value is created, distributed and captured across the ecosystem, giving it a significant competitive advantage.

A key foundation of Google's ecosystem is Android, which powers a large share of smartphones globally. Although hardware is produced by various manufacturers, Android provides the operating system layer that connects users to Google's services such as Search, Maps, Gmail and the Play Store. Search further strengthens the ecosystem by serving as a primary gateway to information, generating valuable user insights that help improve services and support continuous innovation.

Google's advertising business acts as the economic engine of the ecosystem. By connecting advertisers with users across Search, YouTube and partner platforms, Google generates revenue that funds infrastructure development, product enhancement and AI innovation. The combination of user engagement, behavioural data and monetisation creates a powerful feedback loop that continuously strengthens the ecosystem and expands its capabilities.

AI has become a critical layer that enhances Google's entire ecosystem. Generative AI capabilities are increasingly integrated across Search, Android, Workspace and Google Cloud, improving user experiences and creating new opportunities for businesses and developers. Google's ecosystem strength is further reinforced by extensive data networks, strong platform integration and a large developer community. Together, these elements demonstrate how a dominant focal firm can orchestrate a highly interconnected ecosystem that drives innovation, scalability and long-term competitive advantage.

Video 6: Ecosystems Without a Strong Focal Firm

Ecosystems without a strong focal firm operate differently from those controlled by a dominant organisation. Instead of relying on a single company to coordinate activities, these ecosystems function through shared infrastructure, common standards and collaboration among multiple participants. In such environments, value is created collectively by banks, technology providers, regulators, businesses and users. This model demonstrates that large-scale digital innovation can occur without a single organisation controlling the entire ecosystem.

A strong example of this approach is India's Unified Payments Interface (UPI). UPI operates on shared digital infrastructure that enables banks, fintech companies and payment applications to interact seamlessly. Multiple providers compete for customers while relying on the same underlying payment network. Because access is based on common standards and interoperability, new participants can enter the ecosystem more easily, encouraging innovation and expanding consumer choice.

Decentralised ecosystems are also common in sectors such as logistics and digital commerce, where multiple organisations collaborate to deliver services. Retailers, transport providers, warehouse operators and technology vendors work together through shared systems and data exchanges. AI-powered tools for route optimisation, demand forecasting and operational planning can operate across these interconnected networks, allowing innovation to emerge from collaboration between many participants rather than from a single orchestrating firm.

These ecosystems are characterised by interoperability, distributed governance and shared decision-making. Open standards and APIs allow systems to connect efficiently, while regulators, industry bodies and participating organisations collectively establish rules and oversight. Although this approach can create challenges related to coordination and governance, it also reduces dependence on a single provider and encourages broader participation across the ecosystem.

In the context of Generative and Agentic AI, decentralised ecosystems provide an alternative model for innovation and competition. AI capabilities can be integrated across multiple organisations, platforms and services through shared infrastructure and standards. While managing such ecosystems may require greater coordination among stakeholders, they demonstrate how collaboration, interoperability and distributed intelligence can drive large-scale value creation in the digital economy.

Video 7: AI-Driven Growth of Smaller Ecosystems

For many years, building a successful digital ecosystem required significant investment in technology, infrastructure and specialised expertise. This often made it difficult for smaller organisations to compete with large platform companies that possessed greater resources and scale. AI is changing this landscape by lowering entry barriers and making advanced capabilities more accessible. As a result, smaller firms can now build specialised ecosystems, innovate more quickly and compete more effectively in niche markets.

One of the key drivers of this shift is the availability of AI through cloud platforms, APIs and open models. Organisations no longer need to develop complex AI systems from the ground up. Instead, they can access powerful tools and pre-trained models, customise them for specific use cases and deploy solutions rapidly. This significantly reduces development costs and enables smaller businesses to focus on their domain expertise and customer needs rather than building technical infrastructure.

AI is also helping smaller ecosystems scale more efficiently. Cloud-based services allow organisations to increase computing resources as demand grows without investing in physical infrastructure. This flexibility enables startups and small businesses to offer advanced AI-driven solutions, such as analytics, automation and decision support, while remaining agile and cost-effective. The ability to scale on demand makes growth more accessible and sustainable.

Beyond accessibility and scalability, AI improves operational efficiency through automation and intelligent decision-making. Organisations can optimise processes, predict outcomes and use data more effectively across areas such as manufacturing, retail and business services. By reducing costs

and improving productivity, AI strengthens the viability of smaller ecosystems and enables participants to create value collaboratively. In the era of Generative and Agentic AI, ecosystem participation is no longer limited to large enterprises, allowing smaller organisations to innovate, integrate and compete with greater speed and agility.

Video 8: Future of Ecosystems: Co-opetition & Strategic Collaboration

Co-opetition, a combination of cooperation and competition, is becoming a defining feature of modern business ecosystems. As AI technologies become more complex and resource-intensive, organisations increasingly rely on shared infrastructure, standards and platforms while continuing to compete in the marketplace. Rather than operating in isolation, companies collaborate on foundational capabilities and differentiate themselves through products, services, customer experience and innovation. This shift is particularly evident in the era of Generative and Agentic AI, where ecosystem participation is often essential for success.

A clear example of co-opetition can be seen in industries such as telecommunications, where competing organisations rely on shared infrastructure while competing for customers and market share. Similar patterns are emerging across AI-driven sectors, where businesses use common cloud platforms, AI tools and technology frameworks. Shared infrastructure helps reduce costs, accelerate development and create opportunities for innovation, while competition remains focused on delivering unique value to customers.

Industry collaborations are also becoming increasingly important. Organisations, research institutions and technology providers are working together to develop shared AI capabilities, standards and frameworks that benefit entire sectors. At the same time, participating firms continue to compete by applying these capabilities in different ways and creating specialised solutions for their customers. This balance between collaboration and competition helps strengthen ecosystems while encouraging innovation and market differentiation.

As ecosystems become more interconnected, organisations must think strategically about their position within them. Success depends not only on developing strong products but also on building systems that integrate effectively with partners and platforms, avoiding excessive dependence on any single provider and identifying the role they want to play within the ecosystem. Ultimately, the future of business ecosystems will be shaped by organisations that can collaborate where it creates collective value, compete where it drives differentiation and leverage ecosystem intelligence to create sustainable competitive advantage.

Video 9: Final Thoughts

This module highlighted how competition is increasingly shaped by ecosystems rather than individual organisations. Businesses now create value through interconnected networks of platforms, partners, developers and users. As a result, success depends on how effectively organisations participate in and position themselves within these ecosystems.

We also explored how AI is accelerating ecosystem growth by enabling faster innovation, stronger data-driven intelligence and easier access to advanced technologies. Examples such as Google’s ecosystem, UPI’s shared infrastructure and emerging AI-driven ecosystems demonstrated how different models can support innovation and value creation.

Finally, we examined the rise of co-opetition, where organisations collaborate and compete simultaneously. In the era of Generative and Agentic AI, organisations that embrace interoperability, build strategic partnerships and leverage ecosystem intelligence will be best positioned to achieve sustainable growth and long-term competitive advantage.